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FACULTY OF SCIENCE AND TECHNOLOGY
S.E (E&TC / ETX / EC) (NEP) Sem - III
Examination May/June-2026
Electronics Devices & Circuits

[Time: 3:00 Hours]**[Max. Marks: 60]**

N. B

Please check whether you have got the right question paper.

- 1) Part A: Question No 1 is compulsory.
- 2) Attempt any four questions from remaining Part B: Q.2 to Q.7.

PART-A**Q.1 Solve any 10 following questions:****20**

1. Push-pull amplifier reduces:
 - A) Noise
 - B) Distortion
 - C) Voltage
 - D) Current
2. A filter circuit is used to:
 - A) Increase voltage
 - B) Remove AC components
 - C) Amplify signal
 - D) Generate signal
3. Peak Inverse Voltage (PIV) is:
 - A) Forward voltage
 - B) Reverse breakdown voltage
 - C) Maximum reverse voltage a diode can withstand
 - D) Average voltage
4. In active region, transistor acts as:
 - A) Switch
 - B) Amplifier
 - C) Rectifier
 - D) Filter
5. Emitter follower has:
 - A) High voltage gain
 - B) Low output impedance
 - C) High output impedance
 - D) No gain
6. Positive feedback means:
 - A) Output is subtracted
 - B) Output is in phase with input
 - C) Output is 180° out of phase
 - D) No output
7. If R increases in RC oscillator, frequency:
 - A) Increases
 - B) Decreases
 - C) Remains same
 - D) Zero
8. A clamper circuit is used to:
 - A) Clip signal
 - B) Amplify signal
 - C) Shift DC level
 - D) Filter signal
9. The drain current in JFET is maximum at:
 - A) Cut-off
 - B) Pinch-off
 - C) Breakdown
 - D) Zero bias
10. Small signal amplifier is used for:
 - A) High Power output
 - B) Low power signals
 - C) Only DC
 - D) High voltage only

11. Which oscillator has best stability among LC types?
 A) Hartley B) Colpitts C) Clapp D) RC
12. UJT oscillator is also called:
 A) Harmonic oscillator B) Relaxation oscillator
 C) LC oscillator D) RC oscillator
13. RC integrator acts as:
 A) High-pass filter B) Low-pass filter
 C) Band-pass filter D) Oscillator

PART-B

Solve any Four from the following:

- Q.2** A. Draw and explain a colpitts oscillator? **5**
 B. Explain Push - Pull amplifier with advantages and disadvantages? **5**
- Q.3** A. Explain how MOSFET work as : **5**
 i) An Amplifier
 ii) As A switch.
 B. Explain frequency response of amplifier with graph? **5**
- Q.4** A. Compare BJT, JFET, and MOSFET? **5**
 B. Explain with neat circuit diagram, explain the operation of RC phase shift oscillator? **5**
- Q.5** A. Define : Q-point, ripple factor, biasing, clipper and clamper? **5**
 B. Draw and explain transistor biasing methods? **5**
- Q.6** A. Explain the circuit of double tune amplifier in detail? **5**
 B. Explain half wave and full wave rectifier? **5**
- Q.7** A. State the Barkhusain Criteria Oscillation **5**
 B. State advantages and disadvantages of negative feedback **5**